

Karan Anchan

M.Sc. Computer Science (AI) | ML Research & Engineering

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Portfolio: <https://karan-anchan.github.io>

Residence Permit: §16b Student (Valid until 21.01.2028)



EDUCATION

M.Sc. Computer Science (Artificial Intelligence) Albert Ludwig University of Freiburg (ALU) Focus: Deep Learning, Probabilistic Graphical Models, Statistical Pattern Recognition, Robot Mechanics.	Freiburg, Germany <i>Apr 2025 – Present</i>
B.E. Computer Science N.M.A.M Institute of Technology GPA: 9.33/10 (German Equivalent: 1,3). Coursework: Software Engineering, Algorithms, Machine Learning.	Karkala, India <i>Sep 2020 – Aug 2024</i>

WORK EXPERIENCE

Machine Learning Intern WiZdom Ed - Built a LangChain and ChromaDB RAG pipeline for student support, owning ingestion, recursive chunking, embeddings and retrieval across 5,000+ internal documents. - Evaluated 100 batches, averaging 71.7% Recall@5, 93.4% groundedness and 89.1% refusal accuracy, with 2.72 s batch-level p95 response latency.	Mangalore, India <i>Oct 2023 – Oct 2024</i>
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TECHNICAL PROJECTS

Mamba-2 & Attention Hybrid LM — Matched-Token Ratio Study github.com/Karan-Anchan/mamba-hybrid-lm Interactive Evidence Showcase - Trained three 52–54M-parameter hybrid LMs on 700M matched OpenWebText tokens each; the 1:3 attention-to-SSM ratio reached the best validation perplexity (26.30) and 52.3 tok/s sampled generation on an RTX 5070. - Implemented recurrent 8K inference and a checkpoint-verifying FastAPI/SSE service; 1:15 cut logical state 66.3% at 8K for +0.212 perplexity, with checksummed artifacts and a public React showcase.
Edge-First Object Detection — One Model, Three Runtimes (YOLO26) github.com/Karan-Anchan/edge-yolo26-deployment Live WebGPU Demo - Fine-tuned an NMS-free YOLO26 detector on dense retail shelves (SKU-110K, mAP@50-95 0.572) and shipped one ONNX graph to TensorRT (GPU), ONNX Runtime (CPU) and WebGPU (in-browser). - Benchmarked FP16/FP8/INT8 with MLPerf-style latency and NVML power: FP8 hit 560 FPS, FP16 won at 9.3 FPS/W, and per-channel CPU INT8 with an FP32 head limited accuracy loss to 0.72% versus 5.65% in TensorRT.
Offline-to-Online Reinforcement Learning — RLPD Reproduction & Humanoid Extension github.com/Karan-Anchan/rlpd-offline-to-online-rl - Reproduced RLPD (Ball et al., ICML 2023) from scratch in PyTorch with Minari offline datasets (3-person lab project): SAC with symmetric 50/50 sampling, LayerNorm critics and a high-UTD critic ensemble. - Reached 88–90% of the Minari v5 expert across Hopper, Walker2d and HalfCheetah (3 seeds), then found 6.6% offline-state coverage on Humanoid and a +21.9-point advantage for online-only replay over the 50/50 mix at 500k steps.

CERTIFICATIONS

Machine Learning Operations (MLOps) Duke University	Verify Credential
Machine Learning Specialization Stanford & DeepLearning.AI	Verify Credential

SKILLS

- Programming & Dev:** Python, C++, C, SQL, TypeScript/JavaScript, Git, Docker, CI/CD.
- ML & Research:** PyTorch, MuJoCo/Gymnasium, Minari, MONAI, Transformers, LangChain, W&B.
- Inference & Apps:** ONNX Runtime, TensorRT, WebGPU, FastAPI/SSE, React, Vite.
- Languages:** English (C2/Native), German (A2/Pursuing B1), Hindi (Native).